calculate_preTARE_am_kpis_sparkGap_COP

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Pre-TARE Adoption KPIs: Spark Gap, Thermal COP, Break-Even COP

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Computes adoption economics metrics that can be calculated directly from EUSS data and EIA fuel prices, without requiring a TARE model run.

Workflow: Load EUSS data → spark gap → thermal COP & AFUE → break-even COP → maps

Batch Mode: Set `input_measure_package` before calling via `%run -i` to skip prompts.

See `README_adoption_kpis.md` for methodology notes and design decisions.

Step 0: Imports and Configuration

[1]:
```
import os
import pandas as pd
import numpy as np
import geopandas as gpd
import matplotlib.pyplot as plt
import matplotlib.colors as mcolors

from config import PROJECT_ROOT
from cmu_tare_model.constants import ALLOWED_HOUSING_TYPES, VALID_MENU_MPS,␣
 ↪VERBOSE

from cmu_tare_model.adoption_kpis.kpi_functions import (
    mp_to_upgrade,
    load_euss_baseline,
    load_euss_upgrade,
    calculate_price_ratios,
    compute_thermal_cop,          # renamed from compute_thermal_cop_by_state
    compute_breakeven_cop,
    compute_spark_gap_metrics,
    iecc_to_cz_group,             # new - climate zone mapping
    COP_BENCHMARK_RANGES,         # new - validation benchmark ranges
    FUEL_PRICES_PATH,
    SHAPEFILE_PATH,
```

```
    HEATING_LOAD_COL,
    CLIMATE_ZONE_COL,              # new - column name constant
)
from cmu_tare_model.adoption_kpis.visualize_geospatial_data import (
    prepare_state_geodataframe,
    create_choropleth_map,
)

pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 60)

print("  Imports loaded")
print(f"Project root: {PROJECT_ROOT}")
```

```
Project root directory: c:\users\jorda\desktop\projects\cmu-tare-model
  Imports loaded
Project root: c:\users\jorda\desktop\projects\cmu-tare-model
```

Step 0b: Measure Package Selection

[2]:
```
SELECTABLE_MPS = [mp for mp in VALID_MENU_MPS if mp != 0]

try:
    _ = input_measure_package
    batch_mode = True
    selected_mps = [int(input_measure_package)]
    print(f"BATCH MODE: Running for MP{selected_mps[0]}")
except NameError:
    batch_mode = False
    print(f"Available measure packages: {SELECTABLE_MPS}")
    mp_input = input("Enter MP numbers (comma-separated, or 'all'): ").strip()
    if mp_input.lower() == 'all':
        selected_mps = SELECTABLE_MPS
    else:
        selected_mps = [int(x.strip()) for x in mp_input.split(',') if x.
 ↪strip().isdigit()]
        selected_mps = [mp for mp in selected_mps if mp in SELECTABLE_MPS]
    if not selected_mps:
        selected_mps = [4]
        print("No valid MPs selected. Defaulting to MP4.")

print(f"\nSelected measure packages: {selected_mps}")
```

```
Available measure packages: [3, 4, 8]

Selected measure packages: [3, 4]
```

Step 1: Load EUSS Data

[3]:
```
print("=" * 80)
print("STEP 1: LOAD EUSS DATA")
print("=" * 80)

df_baseline = load_euss_baseline()
print(f"  Baseline: {len(df_baseline):,} occupied SF homes")

upgrade_data = {}
for mp in selected_mps:
    upgrade_name = mp_to_upgrade(mp)
    print(f"\nLoading MP{mp} ({upgrade_name})...")
    upgrade_data[mp] = load_euss_upgrade(upgrade_name)
    print(f"  MP{mp}: {len(upgrade_data[mp]):,} applicable homes")

print(f"\n  STEP 1 COMPLETE")
```

```
================================================================================
STEP 1: LOAD EUSS DATA
================================================================================
Loading baseline from: c:\users\jorda\desktop\projects\cmu-tare-model\cmu_tare_m
odel\data\euss_data\resstock_amy2018_release_1.1\national\csv\baseline_metadata_
and_annual_results.csv
  After occupancy filter: 482,597 / 548,916
  After housing type filter (['Single-Family Attached', 'Single-Family
Detached']): 331,531
  Baseline: 331,531 occupied SF homes

Loading MP3 (upgrade03)…
Loading upgrade03 from: c:\users\jorda\desktop\projects\cmu-tare-model\cmu_tare_
model\data\euss_data\resstock_amy2018_release_1.1\national\csv\upgrade03_metadat
a_and_annual_results.csv
  After occupancy filter: 482,597 / 548,916
  After housing type filter: 331,531
  After applicability filter: 331,526
  MP3: 331,526 applicable homes

Loading MP4 (upgrade04)…
Loading upgrade04 from: c:\users\jorda\desktop\projects\cmu-tare-model\cmu_tare_
model\data\euss_data\resstock_amy2018_release_1.1\national\csv\upgrade04_metadat
a_and_annual_results.csv
  After occupancy filter: 482,597 / 548,916
  After housing type filter: 331,531
  After applicability filter: 331,526
  MP4: 331,526 applicable homes
```

```
STEP 1 COMPLETE
```

Step 2: Fuel Price Ratios (Spark Gap)

[4]:
```
print("===== STEP 2: FUEL PRICE RATIOS (CSV, 2024 nominal) =====")
df_prices_csv = calculate_price_ratios(FUEL_PRICES_PATH, year=2024)
print(f"  Loaded price data for {len(df_prices_csv)} states")

print(f"\nSpark Gap - Mean: {df_prices_csv['spark_gap'].mean():.2f}, "
      f"Range: {df_prices_csv['spark_gap'].min():.2f}-
{df_prices_csv['spark_gap'].max():.2f}")

print(f"\n--- Top 5 (highest spark gap) ---")
print(df_prices_csv.head(5).to_string(index=False))
print(f"\n--- Bottom 5 (lowest spark gap) ---")
print(df_prices_csv.tail(5).to_string(index=False))

# TODO: Load updated 10-year average residential fuel price data when available
print("\n  10-year average prices: PLACEHOLDER - awaiting updated data")
```

```
===== STEP 2: FUEL PRICE RATIOS (CSV, 2024 nominal) =====
  Loaded price data for 51 states

Spark Gap - Mean: 3.55, Range: 1.70-6.44

--- Top 5 (highest spark gap) ---
state  state_name  elec_price_kwh  gas_price_kwh  elec_price_mmbtu
gas_price_mmbtu  spark_gap
   AK      Alaska          0.2482         0.0386             72.74
11.30       6.44
   MI    Michigan          0.1930         0.0354             56.56
10.37       5.46
   CT Connecticut          0.2875         0.0553             84.26
16.21       5.20
   WI   Wisconsin          0.1718         0.0333             50.35
9.77        5.15
   CA  California          0.3197         0.0629             93.69
18.44       5.08

--- Bottom 5 (lowest spark gap) ---
state state_name  elec_price_kwh  gas_price_kwh  elec_price_mmbtu
gas_price_mmbtu  spark_gap
   MO   Missouri          0.1291         0.0554             37.84
16.24       2.33
   AR   Arkansas          0.1232         0.0529             36.11
15.50       2.33
   AZ    Arizona          0.1491         0.0673             43.70
```

```
19.72       2.22
   LA  Louisiana          0.1173         0.0538             34.38
15.78       2.18
   FL    Florida          0.1414         0.0834             41.44
24.44       1.70

  10-year average prices: PLACEHOLDER - awaiting updated data
```

Step 3: Thermal COP and Baseline AFUE

[5]:
```python
cop_results = {}

for mp in selected_mps:
    print(f"\n{'=' * 80}")
    print(f"STEP 3: THERMAL COP & BASELINE AFUE (MP{mp}, Natural Gas homes)")
    print(f"{'=' * 80}")

    cop_results[mp] = compute_thermal_cop(
        df_baseline, upgrade_data[mp],
        group_cols=['state'],
        fuel_filter='Natural Gas',
        verbose=True,
    )

    df_cop_mp = cop_results[mp]
    print(f"\n--- Top 5 by COP (MP{mp}) ---")
    print(df_cop_mp.sort_values('thermal_cop', ascending=False).head(5)[
        ['state', 'thermal_cop', 'baseline_afue', 'fans_pumps_pct',␣
↪'home_count']
    ].to_string(index=False))

    print(f"\n--- Bottom 5 by COP (MP{mp}) ---")
    print(df_cop_mp.sort_values('thermal_cop', ascending=True).head(5)[
        ['state', 'thermal_cop', 'baseline_afue', 'fans_pumps_pct',␣
↪'home_count']
    ].to_string(index=False))

    # Validation: heating load consistency
    common_ids = df_baseline.index.intersection(upgrade_data[mp].index)
    Q_baseline = df_baseline.loc[common_ids, HEATING_LOAD_COL].fillna(0)
    Q_upgrade = upgrade_data[mp].loc[common_ids, HEATING_LOAD_COL].fillna(0)
    mask = Q_baseline > 0
    pct_diff = ((Q_upgrade[mask] - Q_baseline[mask]) / Q_baseline[mask]).
↪median()
    print(f"\nHeating load consistency: {pct_diff:.1%} median difference␣
↪(expect ±5-10%)")
```

```python
primary_mp = selected_mps[0]
df_cop = cop_results[primary_mp]
df_upgrade_primary = upgrade_data[primary_mp]
print(f"\n STEP 3 COMPLETE - Primary MP: {primary_mp}")
```

```
================================================================================
STEP 3: THERMAL COP & BASELINE AFUE (MP3, Natural Gas homes)
================================================================================
Matched homes (baseline  upgrade): 331,526
Filtered to 'Natural Gas': 180,939 / 331,526 homes (54.6%)
Excluded 683 homes with zero baseline heating (180,256 remaining)

--- Thermal COP Summary (Natural Gas, by state) ---
Groups: 49
Mean:   2.21
Median: 2.20
Range:  1.53 - 3.05
  All groups within expected COP range (1.5-5.0)

--- Baseline AFUE Summary (Natural Gas) ---
Mean:   0.80
Median: 0.80
Range:  0.75 - 0.82

--- Fan/Pump Energy as % of Total HP Electricity ---
Mean:   3.8%
Range:  1.7% - 5.8%

--- Top 5 by COP (MP3) ---
state  thermal_cop  baseline_afue  fans_pumps_pct  home_count
   CA     3.054755       0.801801        4.001501       24571
   FL     2.893052       0.794453        5.682569        1084
   LA     2.758595       0.799742        5.761571        2139
   TX     2.748806       0.804379        5.035227       12362
   DC     2.715392       0.752035        4.055966         338

--- Bottom 5 by COP (MP3) ---
state  thermal_cop  baseline_afue  fans_pumps_pct  home_count
   ND     1.527465       0.803502        1.730071         402
   MN     1.589963       0.801746        1.937747        4687
   ME     1.647215       0.819947        1.661832          61
   SD     1.673837       0.797856        2.435159         520
   WI     1.699153       0.804310        2.207530        4759

Heating load consistency: 0.1% median difference (expect ±5-10%)
```

```
================================================================================
STEP 3: THERMAL COP & BASELINE AFUE (MP4, Natural Gas homes)
================================================================================
Matched homes (baseline  upgrade): 331,526
Filtered to 'Natural Gas': 180,939 / 331,526 homes (54.6%)
Excluded 683 homes with zero baseline heating (180,256 remaining)

--- Thermal COP Summary (Natural Gas, by state) ---
Groups: 49
Mean:   3.89
Median: 3.98
Range:  2.15 - 5.72
  8 groups with suspicious COP (<1.5 or >5.0):
    AL: 5.06
    CA: 5.72
    FL: 5.70
    GA: 5.06
    LA: 5.35
    MS: 5.02
    SC: 5.06
    TX: 5.30

--- Baseline AFUE Summary (Natural Gas) ---
Mean:   0.80
Median: 0.80
Range:  0.75 - 0.82

--- Fan/Pump Energy as % of Total HP Electricity ---
Mean:   4.5%
Range:  2.4% - 5.7%

--- Top 5 by COP (MP4) ---
state  thermal_cop  baseline_afue  fans_pumps_pct  home_count
   CA     5.719542       0.801801        5.075078       24571
   FL     5.697018       0.794453        5.459375        1084
   LA     5.345519       0.799742        5.496742        2139
   TX     5.303021       0.804379        5.558138       12362
   AL     5.058210       0.805110        5.564781        1853

--- Bottom 5 by COP (MP4) ---
state  thermal_cop  baseline_afue  fans_pumps_pct  home_count
   ND     2.149135       0.803502        2.371268         402
   MN     2.359931       0.801746        2.772684        4687
   ME     2.444452       0.819947        2.846700          61
   SD     2.502694       0.797856        3.040386         520
   WI     2.623592       0.804310        3.174381        4759

Heating load consistency: 5.2% median difference (expect ±5-10%)
```

```
STEP 3 COMPLETE - Primary MP: 3
```

Task B: COP Direction Check (Zero-Baseline Filter Validation)

Validates that the zero-baseline filter (excluding homes with no prior heating system) lowers COP as expected. Compares fixed (default) vs unfixed (`require_baseline_heating=False`) results per state.

[6]:
```python
print("=" * 80)
print("TASK B: COP DIRECTION CHECK (zero-baseline filter validation)")
print("=" * 80)

for mp in selected_mps:
    print(f"\n--- MP{mp} ---")

    # Unfixed: include homes with zero baseline heating (old behavior)
    cop_unfixed = compute_thermal_cop(
        df_baseline, upgrade_data[mp],
        group_cols=['state'],
        fuel_filter='Natural Gas',
        require_baseline_heating=False,
        verbose=False,
    )

    # Fixed: cop_results[mp] already computed with␣
 ↪require_baseline_heating=True (default)
    cop_fixed = cop_results[mp]

    # Merge for comparison
    df_compare = cop_unfixed[['state', 'thermal_cop']].rename(
        columns={'thermal_cop': 'cop_unfixed'}
    ).merge(
        cop_fixed[['state', 'thermal_cop']].rename(
            columns={'thermal_cop': 'cop_fixed'}
        ),
        on='state', how='outer',
    )
    df_compare['delta'] = df_compare['cop_fixed'] - df_compare['cop_unfixed']
    df_compare['direction_ok'] = df_compare['cop_fixed'] <=␣
 ↪df_compare['cop_unfixed']

    print(f"\n{'State':<6} {'COP_unfixed':>12} {'COP_fixed':>10} {'Delta':>8}␣
 ↪{'Dir OK?':>8}")
    print("-" * 48)
    for _, r in df_compare.sort_values('state').iterrows():
```

```
        print(f"{r['state']:<6} {r['cop_unfixed']:>12.3f} {r['cop_fixed']:>10.
↪3f} "
              f"{r['delta']:>+8.3f} {' ' if r['direction_ok'] else ' ':>8}")

    n_pass = df_compare['direction_ok'].sum()
    n_total = len(df_compare)
    n_fail = n_total - n_pass
    print(f"\nSummary: {n_pass}/{n_total} states pass direction check")
    if n_fail > 0:
        fails = df_compare[~df_compare['direction_ok']]
        print(f"  {n_fail} state(s) where fixed COP > unfixed (investigate):")
        for _, r in fails.iterrows():
            print(f"    {r['state']}: unfixed={r['cop_unfixed']:.3f}, "
                  f"fixed={r['cop_fixed']:.3f}, delta={r['delta']:+.3f}")
    else:
        print("  All states: fixed COP  unfixed COP (filter lowered COP as␣
↪expected)")

print(f"\n  TASK B COMPLETE")
```

```
================================================================================
TASK B: COP DIRECTION CHECK (zero-baseline filter validation)
================================================================================

--- MP3 ---

State   COP_unfixed  COP_fixed    Delta  Dir OK?
-------------------------------------------------
AL            2.608      2.608   +0.000
AR            2.499      2.499   +0.000
AZ            2.587      2.588   +0.001
CA            3.054      3.055   +0.001
CO            1.952      1.952   +0.000
CT            2.086      2.086   +0.000
DC            2.715      2.715   +0.000
DE            2.315      2.315   +0.000
FL            2.893      2.893   +0.000
GA            2.599      2.599   +0.000
IA            1.763      1.763   +0.000
ID            2.155      2.155   +0.000
IL            1.946      1.946   +0.000
IN            1.972      1.972   +0.000
KS            2.201      2.201   +0.000
KY            2.290      2.290   +0.000
LA            2.759      2.759   +0.000
MA            2.012      2.012   +0.000
MD            2.399      2.399   +0.000
```

```
ME            1.647      1.647   +0.000
MI            1.786      1.786   +0.000
MN            1.590      1.590   +0.000
MO            2.228      2.228   +0.000
MS            2.584      2.584   +0.000
MT            1.892      1.892   +0.000
NC            2.471      2.471   +0.000
ND            1.527      1.527   +0.000
NE            1.898      1.898   +0.000
NH            1.854      1.854   +0.000
NJ            2.294      2.294   +0.000
NM            2.384      2.384   +0.000
NV            2.491      2.491   +0.001
NY            1.974      1.974   +0.000
OH            1.990      1.990   +0.000
OK            2.493      2.493   +0.000
OR            2.619      2.619   +0.000
PA            2.020      2.020   +0.000
RI            1.937      1.937   +0.000
SC            2.603      2.603   +0.000
SD            1.674      1.674   +0.000
TN            2.408      2.408   +0.000
TX            2.749      2.749   +0.000
UT            2.163      2.163   +0.000
VA            2.380      2.380   +0.000
VT            1.731      1.731   +0.000
WA            2.510      2.510   +0.000
WI            1.699      1.699   +0.000
WV            2.051      2.051   +0.000
WY            1.977      1.977   +0.000

Summary: 44/49 states pass direction check
  5 state(s) where fixed COP > unfixed (investigate):
    AZ: unfixed=2.587, fixed=2.588, delta=+0.001
    CA: unfixed=3.054, fixed=3.055, delta=+0.001
    FL: unfixed=2.893, fixed=2.893, delta=+0.000
    NV: unfixed=2.491, fixed=2.491, delta=+0.001
    TX: unfixed=2.749, fixed=2.749, delta=+0.000

--- MP4 ---

State   COP_unfixed  COP_fixed    Delta  Dir OK?
-------------------------------------------------
AL            5.058      5.058   +0.000
AR            4.830      4.830   +0.000
AZ            4.685      4.685   -0.000
CA            5.720      5.720   -0.000
CO            3.075      3.075   +0.000
```

```
CT              3.595      3.595   +0.000
DC              4.981      4.981   +0.000
DE              4.261      4.261   +0.000
FL              5.697      5.697   +0.000
GA              5.057      5.057   +0.000
IA              2.850      2.850   +0.000
ID              3.594      3.594   +0.000
IL              3.302      3.302   +0.000
IN              3.337      3.337   +0.000
KS              3.980      3.980   +0.000
KY              4.146      4.146   +0.000
LA              5.346      5.346   +0.000
MA              3.375      3.375   +0.000
MD              4.428      4.428   +0.000
ME              2.444      2.444   +0.000
MI              2.829      2.829   +0.000
MN              2.360      2.360   +0.000
MO              4.083      4.083   +0.000
MS              5.024      5.024   +0.000
MT              2.846      2.846   +0.000
NC              4.724      4.724   +0.000
ND              2.149      2.149   +0.000
NE              3.180      3.180   +0.000
NH              3.056      3.056   +0.000
NJ              4.131      4.131   +0.000
NM              4.107      4.107   +0.000
NV              4.481      4.481   -0.000
NY              3.202      3.202   +0.000
OH              3.343      3.343   +0.000
OK              4.800      4.800   +0.000
OR              4.661      4.661   +0.000
PA              3.342      3.342   +0.000
RI              3.167      3.167   +0.000
SC              5.057      5.057   +0.000
SD              2.503      2.503   +0.000
TN              4.567      4.567   +0.000
TX              5.303      5.303   +0.000
UT              3.698      3.698   +0.000
VA              4.460      4.460   +0.000
VT              2.676      2.676   +0.000
WA              4.220      4.220   +0.000
WI              2.624      2.624   +0.000
WV              3.436      3.436   +0.000
WY              2.960      2.960   +0.000

Summary: 49/49 states pass direction check
  All states: fixed COP   unfixed COP (filter lowered COP as expected)
```

```
TASK B COMPLETE
```

Task C: Climate Zone Benchmark Validation

Aggregates thermal COP by IECC climate zone group and validates against literature-derived benchmark ranges (`COP_BENCHMARK_RANGES`). Also produces a state × climate zone cross-tab with a PA spot check.

[7]:
```python
print("=" * 80)
print("TASK C: CLIMATE ZONE BENCHMARK VALIDATION")
print("=" * 80)

for mp in selected_mps:
    mp_key = f'mp{mp}'
    print(f"\n{' ' * 60}")
    print(f"MP{mp} - Climate Zone Group Aggregation")
    print(f"{' ' * 60}")

    # --- C1: COP by climate zone group ---
    df_cop_cz = compute_thermal_cop(
        df_baseline, upgrade_data[mp],
        group_cols=['cz_group'],
        fuel_filter='Natural Gas',
        verbose=True,
    )

    # --- C2: COP by state × climate zone group ---
    df_cop_state_cz = compute_thermal_cop(
        df_baseline, upgrade_data[mp],
        group_cols=['state', 'cz_group'],
        fuel_filter='Natural Gas',
        verbose=True,
    )

    # --- C3: Validate against COP_BENCHMARK_RANGES ---
    print(f"\n--- Benchmark Validation (MP{mp}) ---")
    print(f"{'CZ Group':<10} {'Label':<18} {'COP':>6} {'Expected':>14} {'Pass':
↪>6}")
    print("-" * 58)
    n_checked = 0
    n_passed = 0
    for cz_grp, bench in COP_BENCHMARK_RANGES.items():
        row = df_cop_cz[df_cop_cz['cz_group'] == cz_grp]
        if len(row) == 0:
            print(f"{cz_grp:<10} {bench['label']:<18} {'N/A':>6} {'':<14}␣
↪{'SKIP':>6}")
            continue
```

```
        cop_val = row['thermal_cop'].values[0]
        if mp_key in bench:
            lo, hi = bench[mp_key]
            in_range = lo <= cop_val <= hi
            n_checked += 1
            n_passed += int(in_range)
            range_str = f"[{lo:.1f}, {hi:.1f}]"
            print(f"{cz_grp:<10} {bench['label']:<18} {cop_val:>6.2f}␣
↪{range_str:>14} "
                  f"{' ' if in_range else ' ':>6}")
        else:
            print(f"{cz_grp:<10} {bench['label']:<18} {cop_val:>6.2f} {'no␣
↪bench':>14} {'-':>6}")

    print(f"\nOverall: {n_passed}/{n_checked} climate zone groups within␣
↪expected range")

    # --- C4: PA spot check ---
    pa_rows = df_cop_state_cz[df_cop_state_cz['state'] == 'PA']
    if len(pa_rows) > 0:
        print(f"\n--- PA Spot Check (MP{mp}) ---")
        # ASSUMPTION: PA is primarily CZ 4-5
        pa_ranges = {'mp3': (1.8, 2.4), 'mp4': (2.5, 3.4)}
        for _, pa_row in pa_rows.iterrows():
            cop_val = pa_row['thermal_cop']
            cz = pa_row['cz_group']
            if mp_key in pa_ranges:
                lo, hi = pa_ranges[mp_key]
                ok = lo <= cop_val <= hi
                print(f"  PA (CZ {cz}): COP = {cop_val:.2f}, "
                      f"expected [{lo:.1f}, {hi:.1f}] → {'  PASS' if ok else '␣
↪FAIL'}")
            else:
                print(f"  PA (CZ {cz}): COP = {cop_val:.2f} (no PA benchmark␣
↪for {mp_key})")
    else:
        print("\n  PA not found in state × CZ cross-tab")

print(f"\n  TASK C COMPLETE")
```

```
================================================================================
TASK C: CLIMATE ZONE BENCHMARK VALIDATION
================================================================================


MP3 - Climate Zone Group Aggregation
```

```
Matched homes (baseline  upgrade): 331,526
Filtered to 'Natural Gas': 180,939 / 331,526 homes (54.6%)
Excluded 683 homes with zero baseline heating (180,256 remaining)

--- Thermal COP Summary (Natural Gas, by cz_group) ---
Groups: 3
Mean:   2.15
Median: 2.05
Range:  1.67 - 2.74
  All groups within expected COP range (1.5-5.0)

--- Baseline AFUE Summary (Natural Gas) ---
Mean:   0.80
Median: 0.80
Range:  0.80 - 0.81

--- Fan/Pump Energy as % of Total HP Electricity ---
Mean:   3.4%
Range:  2.2% - 4.8%
Matched homes (baseline  upgrade): 331,526
Filtered to 'Natural Gas': 180,939 / 331,526 homes (54.6%)
Excluded 683 homes with zero baseline heating (180,256 remaining)

--- Thermal COP Summary (Natural Gas, by state × cz_group) ---
Groups: 72
Mean:   2.22
Median: 2.20
Range:  1.53 - 3.15
  All groups within expected COP range (1.5-5.0)

--- Baseline AFUE Summary (Natural Gas) ---
Mean:   0.81
Median: 0.81
Range:  0.75 - 0.85

--- Fan/Pump Energy as % of Total HP Electricity ---
Mean:   3.9%
Range:  1.6% - 6.2%

--- Benchmark Validation (MP3) ---
CZ Group   Label                 COP       Expected   Pass
------------------------------------------------------------
1-3        Warm (CZ 1-3)        2.74     [2.4, 3.2]
4-5        Mixed (CZ 4-5)       2.05     [2.0, 2.8]
6-7        Cold (CZ 6-7)        1.67     [1.6, 2.4]

Overall: 3/3 climate zone groups within expected range
```

```
--- PA Spot Check (MP3) ---
  PA (CZ 4-5): COP = 2.03, expected [1.8, 2.4] →  PASS
  PA (CZ 6-7): COP = 1.63, expected [1.8, 2.4] →  FAIL


MP4 - Climate Zone Group Aggregation

Matched homes (baseline  upgrade): 331,526
Filtered to 'Natural Gas': 180,939 / 331,526 homes (54.6%)
Excluded 683 homes with zero baseline heating (180,256 remaining)

--- Thermal COP Summary (Natural Gas, by cz_group) ---
Groups: 3
Mean:   3.76
Median: 3.47
Range:  2.51 - 5.30
  1 groups with suspicious COP (<1.5 or >5.0):
    1-3: 5.30

--- Baseline AFUE Summary (Natural Gas) ---
Mean:   0.80
Median: 0.80
Range:  0.80 - 0.81

--- Fan/Pump Energy as % of Total HP Electricity ---
Mean:   4.2%
Range:  3.0% - 5.5%
Matched homes (baseline  upgrade): 331,526
Filtered to 'Natural Gas': 180,939 / 331,526 homes (54.6%)
Excluded 683 homes with zero baseline heating (180,256 remaining)

--- Thermal COP Summary (Natural Gas, by state × cz_group) ---
Groups: 72
Mean:   3.89
Median: 3.74
Range:  2.15 - 6.31
  13 groups with suspicious COP (<1.5 or >5.0):
    AL, 1-3: 5.06
    AR, 1-3: 5.02
    AZ, 1-3: 5.82
    CA, 1-3: 5.96
    FL, 1-3: 5.70
    GA, 1-3: 5.06
    GA, 4-5: 5.03
    LA, 1-3: 5.35
    MS, 1-3: 5.02
    NV, 1-3: 6.31
    SC, 1-3: 5.06
```

```
    TX, 1-3: 5.35
    UT, 1-3: 5.35

--- Baseline AFUE Summary (Natural Gas) ---
Mean:   0.81
Median: 0.81
Range:  0.75 - 0.85

--- Fan/Pump Energy as % of Total HP Electricity ---
Mean:   4.5%
Range:  2.4% - 5.8%

--- Benchmark Validation (MP4) ---
CZ Group   Label                 COP       Expected   Pass
------------------------------------------------------------
1-3        Warm (CZ 1-3)        5.30     [3.0, 4.2]
4-5        Mixed (CZ 4-5)       3.47     [2.5, 3.5]
6-7        Cold (CZ 6-7)        2.51     [2.0, 3.0]

Overall: 2/3 climate zone groups within expected range

--- PA Spot Check (MP4) ---
  PA (CZ 4-5): COP = 3.39, expected [2.5, 3.4] →  PASS
  PA (CZ 6-7): COP = 2.28, expected [2.5, 3.4] →  FAIL

  TASK C COMPLETE
```

Task D: Break-Even COP Cross-Validation (Jenkins et al.)

Rich cross-validation against Jenkins Heat Pump Map reference values. For each reference state: compares thermal COP, spark gap, computed break-even COP at 90% AFUE, and Jenkins reference. Determines whether the heat pump beats break-even and whether the computed value matches Jenkins within strict (±0.05) and relaxed (±0.50) tolerances.

Known discrepancy sources: 1. Gas heat content — Jenkins: 1020 BTU/cf; ours: 1038 BTU/cf (EIA average) 2. ~~Price year — Jenkins: earlier year; ours: 2024 EIA nominal~~

[8]:
```
print("=" * 80)
print(f"TASK D: BREAK-EVEN COP CROSS-VALIDATION (Jenkins et al.,␣
  ↪MP{primary_mp})")
print("=" * 80)

# Jenkins Heat Pump Map break-even COP at 90% AFUE reference values
# Source: Jenkins et al.
# ASSUMPTION: Jenkins assumes 1020 BTU/cf gas heat content; we use 1038 BTU/cf
# (current EIA average). This ~1.8% difference propagates into spark gap and
# break-even COP. Our prices are now 2024 EIA nominal (same as Jenkins).
```

```
jenkins_ref = {
    'FL': 1.50, 'PA': 3.51, 'MN': 3.90,
    'AK': 5.69, 'CA': 4.49, 'MA': 3.60,
}

# Use state-level COP from cop_results and spark gap from current prices
df_prices_for_d = calculate_price_ratios(FUEL_PRICES_PATH, year=2024)
df_breakeven_for_d = compute_breakeven_cop(df_prices_for_d, afue_scenarios=[0.
 ↪90])
df_spark_for_d = compute_spark_gap_metrics(
    df_prices_for_d, cop_results[primary_mp],
    df_breakeven=df_breakeven_for_d, verbose=False,
)

header = (f"{'State':<6} {'COP':>6} {'Spark':>6} {'BE_90':>6} "
          f"{'Jenkins':>8} {'Strict':>7} {'Relaxed':>8} {'HP>BE?':>7}")
print(f"\n{header}")
print("-" * len(header))

n_strict = 0
n_relaxed = 0
for st, ref_val in sorted(jenkins_ref.items()):
    row_cop = cop_results[primary_mp][cop_results[primary_mp]['state'] == st]
    row_spark = df_spark_for_d[df_spark_for_d['state'] == st]

    if len(row_cop) == 0 or len(row_spark) == 0:
        print(f"{st:<6} {'N/A':>6} {'N/A':>6} {'N/A':>6} {ref_val:>8.2f} "
              f"{'-':>7} {'-':>8} {'-':>7}")
        continue

    thermal_cop = row_cop['thermal_cop'].values[0]
    spark_gap = row_spark['spark_gap'].values[0]
    # ASSUMPTION: breakeven_cop_90 = spark_gap * 0.90
    breakeven_cop_90 = spark_gap * 0.90

    cop_exceeds = thermal_cop > breakeven_cop_90
    strict = abs(ref_val - breakeven_cop_90) <= 0.05
    relaxed = abs(ref_val - breakeven_cop_90) <= 0.50
    n_strict += int(strict)
    n_relaxed += int(relaxed)

    print(f"{st:<6} {thermal_cop:>6.2f} {spark_gap:>6.2f} {breakeven_cop_90:>6.
 ↪2f} "
          f"{ref_val:>8.2f} {'*' if strict else ' ':>7} "
          f"{'*' if relaxed else ' ':>8} {'Yes' if cop_exceeds else 'No':>7}")

n_total = len(jenkins_ref)
```

```
print(f"\nStrict match (±0.05): {n_strict}/{n_total}")
print(f"Relaxed match (±0.50): {n_relaxed}/{n_total}")
print(f"\nNote: Remaining discrepancies arise from (1) gas heat content "
      f"(Jenkins: 1020, ours: 1038 BTU/cf)")

# Interpretation
print("\n--- Interpretation ---")
warm = ['FL']
cold = ['MN', 'AK']
for label, states in [('Warm', warm), ('Cold', cold)]:
    for st in states:
        row_cop = cop_results[primary_mp][cop_results[primary_mp]['state'] ==␣
 ↪st]
        row_spark = df_spark_for_d[df_spark_for_d['state'] == st]
        if len(row_cop) == 0 or len(row_spark) == 0:
            continue
        cop_val = row_cop['thermal_cop'].values[0]
        be_val = row_spark['spark_gap'].values[0] * 0.90
        exceeds = cop_val > be_val
        msg = "HP saves money on fuel" if exceeds else "HP does NOT beat␣
 ↪furnace on fuel cost"
        print(f"  {st} ({label}): COP={cop_val:.2f}, BE_90={be_val:.2f} →␣
 ↪{msg}")

print(f"\n  TASK D COMPLETE")
```

```
================================================================================
TASK D: BREAK-EVEN COP CROSS-VALIDATION (Jenkins et al., MP3)
================================================================================

State     COP  Spark  BE_90  Jenkins  Strict  Relaxed  HP>BE?
------------------------------------------------------------
AK        N/A    N/A    N/A     5.69       -        -       -
CA       3.05   5.08   4.57     4.49                      No
FL       2.89   1.70   1.53     1.50                     Yes
MA       2.01   4.08   3.67     3.60                      No
MN       1.59   4.41   3.97     3.90                      No
PA       2.02   3.97   3.57     3.51                      No

Strict match (±0.05): 1/6
Relaxed match (±0.50): 5/6

Note: Remaining discrepancies arise from (1) gas heat content (Jenkins: 1020,
ours: 1038 BTU/cf)

--- Interpretation ---
  FL (Warm): COP=2.89, BE_90=1.53 → HP saves money on fuel
```

```
MN (Cold): COP=1.59, BE_90=3.97 → HP does NOT beat furnace on fuel cost

TASK D COMPLETE
```

Step 4: Break-Even COP and Merged Metrics

[9]:
```python
print("=" * 80)
print("STEP 4: BREAK-EVEN COP & MERGED METRICS")
print("=" * 80)

# --- 4a: Compute break-even COP for multiple AFUE scenarios ---
# ASSUMPTION: 2024 EIA state-level residential prices.
afue_scenarios = [0.80, 0.90, 0.95, 1.00]
df_breakeven = compute_breakeven_cop(df_prices_csv,␣
 ↪afue_scenarios=afue_scenarios)
print(f"  Break-even COP computed for {len(df_breakeven)} states, "
      f"AFUE scenarios: {afue_scenarios}")

print(f"\nBreak-even COP @90% AFUE - "
      f"Mean: {df_breakeven['breakeven_cop_90'].mean():.2f}, "
      f"Range: {df_breakeven['breakeven_cop_90'].min():.2f}-"
      f"{df_breakeven['breakeven_cop_90'].max():.2f}")

# --- 4b: Merge prices + COP + break-even into single state-level table ---
df_spark = compute_spark_gap_metrics(
    df_prices_csv, df_cop, df_breakeven=df_breakeven, verbose=True
)

# --- 4c: Validate against Jenkins Heat Pump Map reference values (90% AFUE) ---
# Source: Jenkins et al., break-even COP at 90% AFUE
# Note: Jenkins assumes 1020 BTU/cf heat content (ours: 1038) and a different
# price year. Differences are expected and documented in Task D.
jenkins_ref = {
    'FL': 1.50, 'PA': 3.51, 'MN': 3.90,
    'AK': 5.69, 'CA': 4.49, 'MA': 3.60,
}
print("\n--- Jenkins Validation (break-even COP @90% AFUE, ±0.05) ---")
print(f"{'State':<6} {'Computed':>10} {'Jenkins':>10} {'Diff':>8} {'Pass':>6}")
print("-" * 44)
for st, ref_val in sorted(jenkins_ref.items()):
    row = df_spark[df_spark['state'] == st]
    if len(row) == 0:
        print(f"{st:<6} {'N/A':>10} {ref_val:>10.2f} {'-':>8} {'SKIP':>6}")
        continue
    computed = row['breakeven_cop_90'].values[0]
    diff = computed - ref_val
    passed = abs(diff) <= 0.05
```

```
    print(f"{st:<6} {computed:>10.2f} {ref_val:>10.2f} {diff:>+8.2f} {' ' if␣
 ↪passed else ' ':>6}")

# --- 4d: Cross-validation (effective COP vs break-even COP) ---
print("\n--- Cross-Validation: Effective COP vs Break-Even @90% ---")
warm_states = ['FL', 'GA', 'SC', 'TX', 'LA']
cold_states = ['MN', 'WI', 'VT', 'ND', 'ME']
for label, states in [('Warm', warm_states), ('Cold', cold_states)]:
    subset = df_spark[df_spark['state'].isin(states)].copy()
    if len(subset) == 0:
        continue
    subset['cop_minus_be'] = subset['thermal_cop'] - subset['breakeven_cop_90']
    print(f"\n{label} states:")
    print(f"  {'State':<6} {'COP':>6} {'BE_90':>6} {'Diff':>8} {'HP Wins?':
 ↪>10}")
    for _, r in subset.iterrows():
        hp_wins = r['thermal_cop'] > r['breakeven_cop_90']
        print(f"  {r['state']:<6} {r['thermal_cop']:>6.2f} "
              f"{r['breakeven_cop_90']:>6.2f} {r['cop_minus_be']:>+8.2f} "
              f"{'Yes' if hp_wins else 'No':>10}")

print(f"\n  STEP 4 COMPLETE")
```

```
================================================================================
STEP 4: BREAK-EVEN COP & MERGED METRICS
================================================================================
  Break-even COP computed for 51 states, AFUE scenarios: [0.8, 0.9, 0.95, 1.0]

Break-even COP @90% AFUE - Mean: 3.19, Range: 1.53-5.80

--- Spark Gap Metrics Summary (49 states) ---
Break-even COP @90% AFUE - Mean: 3.16, Range: 1.53-4.91
States where effective COP < break-even @90%: 35/49

--- Jenkins Validation (break-even COP @90% AFUE, ±0.05) ---
State    Computed    Jenkins     Diff   Pass
-------------------------------------------
AK            N/A       5.69        -   SKIP
CA           4.57       4.49    +0.08
FL           1.53       1.50    +0.03
MA           3.67       3.60    +0.07
MN           3.97       3.90    +0.07
PA           3.57       3.51    +0.06

--- Cross-Validation: Effective COP vs Break-Even @90% ---

Warm states:
```

```
  State     COP  BE_90     Diff   HP Wins?
  SC       2.60   2.30    +0.30        Yes
  TX       2.75   2.28    +0.47        Yes
  GA       2.60   2.12    +0.48        Yes
  LA       2.76   1.96    +0.80        Yes
  FL       2.89   1.53    +1.36        Yes

Cold states:
  State     COP  BE_90     Diff   HP Wins?
  WI       1.70   4.64    -2.94         No
  MN       1.59   3.97    -2.38         No
  ME       1.65   3.51    -1.86         No
  ND       1.53   3.50    -1.97         No
  VT       1.73   3.46    -1.73         No

 STEP 4 COMPLETE
```

Step 5: Geospatial Visualization

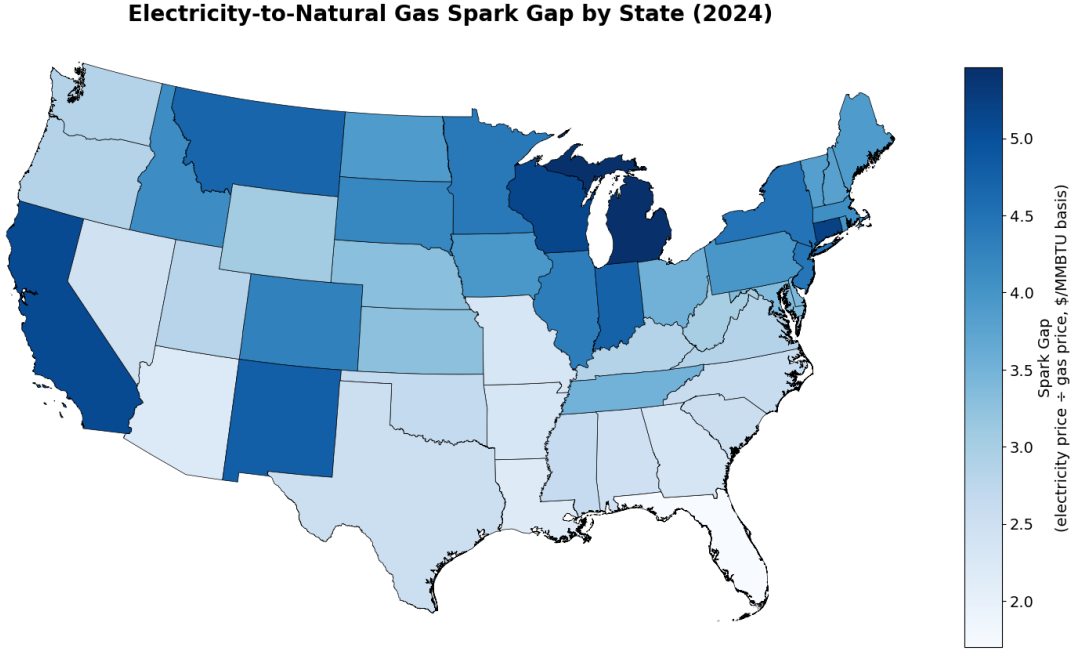
[10]:
```
gdf_conus = None
gdf_alaska = None

try:
    print(f"Loading shapefile from: {SHAPEFILE_PATH}")
    gdf_states_raw = gpd.read_file(SHAPEFILE_PATH)
    _, gdf_conus, gdf_alaska = prepare_state_geodataframe(gdf_states_raw,␣
 ↪df_spark, merge_col='state')
    print(f"  CONUS: {len(gdf_conus)}, Alaska: {len(gdf_alaska)}")
    print(" Geodataframe prepared")
except FileNotFoundError:
    print(f" Shapefile not found at {SHAPEFILE_PATH} - skipping maps")
except Exception as e:
    print(f" Could not load shapefile: {e}")
```

```
Loading shapefile from: c:\users\jorda\desktop\projects\cmu-tare-model\cmu_tare_
model\data\electricity_ng_price_ratio\nhgis0011_shapefile_tl2015_us_state_2015\U
S_state_2015.shp
  CONUS: 49, Alaska: 0
 Geodataframe prepared
```

[11]:
```
if gdf_conus is not None and gdf_alaska is not None:
    print("Generating choropleth maps...")

    # --- Spark gap (blue) ---
    create_choropleth_map(
        gdf_conus, gdf_alaska, column='spark_gap',
        title='Electricity-to-Natural Gas Spark Gap by State (2024)',
```

```
        cbar_label='Spark Gap\n(electricity price ÷ gas price, $/MMBTU basis)',
        output_path=os.path.join(PROJECT_ROOT, "state_spark_gap_map_2024.png"),
        cmap='Blues', show_plot=True,
    )

else:
    print("  Maps skipped - no geodataframe available")
```

```
Generating choropleth maps…
  Saved: c:\users\jorda\desktop\projects\cmu-tare-
model\state_spark_gap_map_2024.png
```



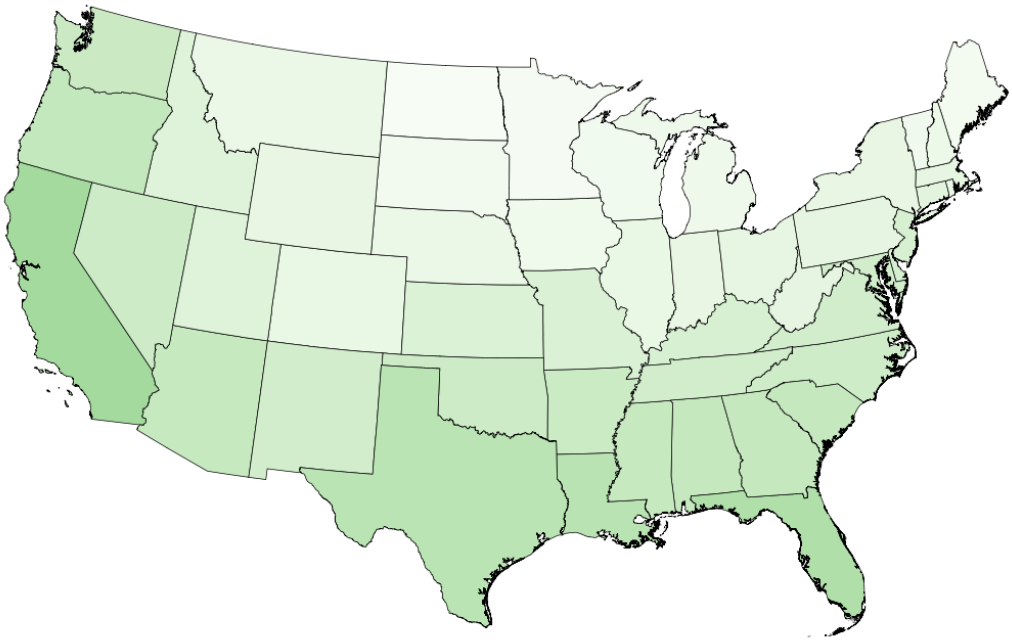
[12]:
```
if gdf_conus is not None and gdf_alaska is not None:
    print("Generating choropleth maps...")

    # --- Thermal COP: MP3 top / MP4 bottom with shared color scale (greens) ---
    cop_mps = [mp for mp in [3, 4] if mp in cop_results]
    if len(cop_mps) >= 2:
        # Build per-MP geodataframes and find shared color range
        gdf_cop_panels = {}
        all_vals = pd.Series(dtype=float)
        for mp_num in cop_mps:
            df_spark_mp = compute_spark_gap_metrics(df_prices_csv,␣
 ↪cop_results[mp_num])
```

```
            _, gdf_c, gdf_a = prepare_state_geodataframe(
                gdf_states_raw, df_spark_mp, merge_col='state'
            )
            gdf_cop_panels[mp_num] = (gdf_c, gdf_a)
            all_vals = pd.concat([all_vals, gdf_c['thermal_cop'],␣
↪gdf_a['thermal_cop']])

        shared_norm = mcolors.Normalize(vmin=all_vals.dropna().min(),␣
↪vmax=all_vals.dropna().max())
        col = 'thermal_cop'
        nrows = len(cop_mps)

        fig, axes_grid = plt.subplots(nrows, 1, figsize=(14, 9 * nrows),␣
↪facecolor='white')
        if nrows == 1:
            axes_grid = [axes_grid]

        for i, mp_num in enumerate(cop_mps):
            gdf_c, gdf_a = gdf_cop_panels[mp_num]
            ax = axes_grid[i]

            gdf_c.plot(ax=ax, column=col, cmap='Greens', edgecolor='black',
                       linewidth=0.5, norm=shared_norm, legend=False)
            ax.set_axis_off()
            ax.set_title(f'MP{mp_num} - Thermal COP by State (2024)',
                         fontsize=18, fontweight='bold', pad=12)

            # Alaska inset - only plot if data exists
            inset_bounds = ax.get_position()
            ax_ak = fig.add_axes([
                inset_bounds.x0, inset_bounds.y0,
                inset_bounds.width * 0.22, inset_bounds.height * 0.28,
            ])
            if not gdf_a.empty:
                gdf_a.plot(ax=ax_ak, column=col, cmap='Greens',␣
↪edgecolor='black',
                           linewidth=0.5, norm=shared_norm, legend=False)
            ax_ak.set_axis_off()

        # Shared colorbar - bottom center
        cax = fig.add_axes([0.25, 0.02, 0.50, 0.015])
        sm = plt.cm.ScalarMappable(cmap='Greens', norm=shared_norm)
        sm.set_array([])
        cbar = fig.colorbar(sm, cax=cax, orientation='horizontal')
        cbar.set_label('Thermal COP (heat delivered / electricity consumed)',␣
↪fontsize=16)
        cax.tick_params(labelsize=16)
```

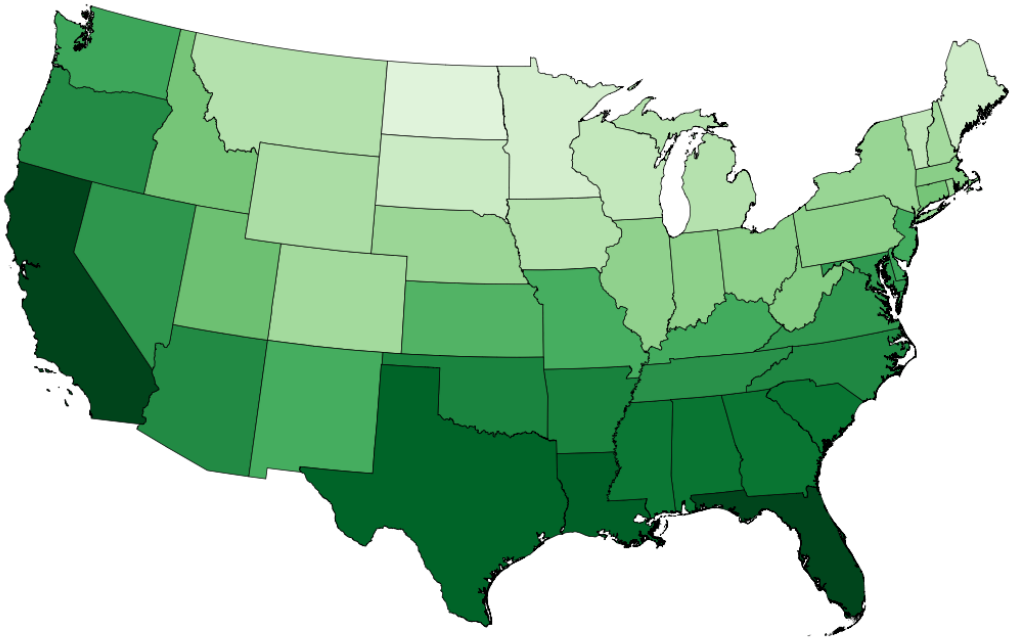
```
        fig.subplots_adjust(hspace=0.04)
        output = os.path.join(PROJECT_ROOT, "state_thermal_cop_mp3_mp4_map_2024.
↪png")
        fig.savefig(output, dpi=600, bbox_inches='tight', facecolor='white')
        print(f"  Saved: {output}")
        plt.show()
    elif len(cop_mps) == 1:
        create_choropleth_map(
            gdf_conus, gdf_alaska, column='thermal_cop',
            title=f'Heat Pump Thermal COP by State (MP{cop_mps[0]}, 2024)',
            cbar_label='Thermal COP\n(heat delivered / electricity consumed)',
            output_path=os.path.join(PROJECT_ROOT,␣
↪f"state_thermal_cop_mp{cop_mps[0]}_map_2024.png"),
            cmap='Greens', show_plot=True,
        )
    else:
        print("  No thermal COP results for MP3 or MP4 - skipping COP maps")

    print("  Maps generated")
else:
    print("  Maps skipped - no geodataframe available")
```

```
Generating choropleth maps…
  Saved: c:\users\jorda\desktop\projects\cmu-tare-
model\state_thermal_cop_mp3_mp4_map_2024.png
```

MP3 — Thermal COP by State (2024)



MP4 — Thermal COP by State (2024)



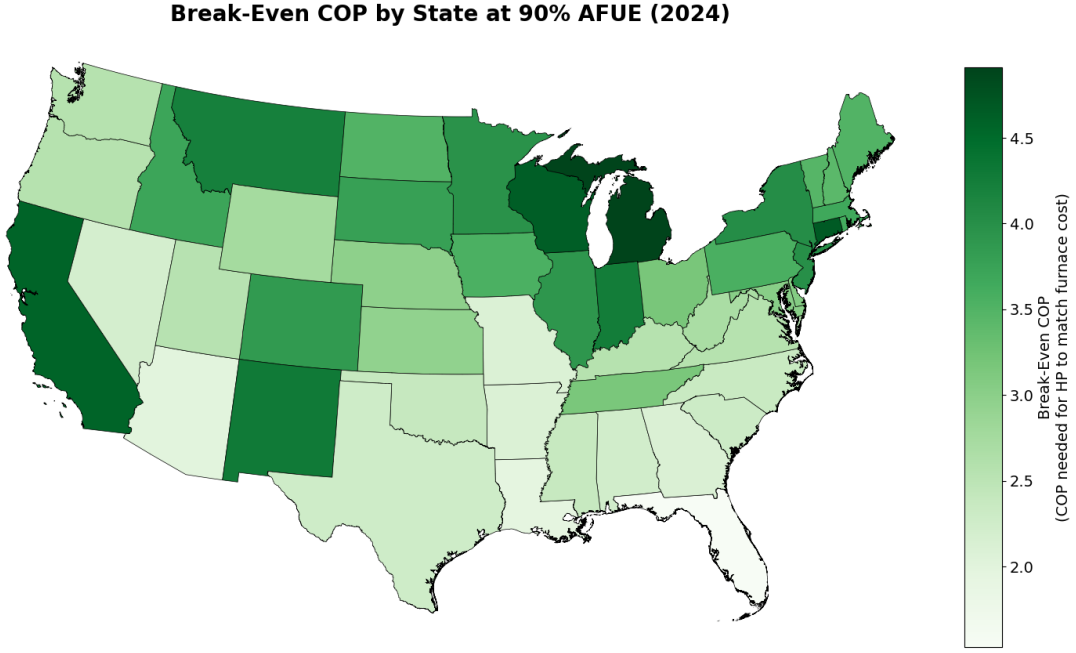
Maps generated

[ ]:

[13]:
```
from typing import Optional, Tuple
import pandas as pd
import numpy as np
import geopandas as gpd
import matplotlib.pyplot as plt
import matplotlib.colors as mcolors

def create_choropleth_map(
    gdf_conus: gpd.GeoDataFrame,
    gdf_alaska: gpd.GeoDataFrame,
    column: str,
    title: str,
    cbar_label: str,
    year: int = 2024,
    output_path:Optional[str] = None,
    figsize: Tuple[int, int] = (16, 10),
    dpi: int = 600,
    cmap: str = 'Blues',
    norm=None,
    show_plot: bool = True
) -> Optional[str]:
    """
    Create a choropleth map with CONUS main panel and Alaska inset.

    Pass a matplotlib Normalize (e.g. TwoSlopeNorm) via `norm` for diverging
    maps; otherwise linear normalization is derived from the data automatically.
    """
    plot_kw = dict(column=column, cmap=cmap, edgecolor='black', linewidth=0.5,␣
 ↪legend=False)

    if norm is not None:
        plot_kw['norm'] = norm
    else:
        vals = pd.concat([gdf_conus[column], gdf_alaska[column]]).dropna()
        norm = mcolors.Normalize(vmin=vals.min(), vmax=vals.max())
        plot_kw['vmin'], plot_kw['vmax'] = vals.min(), vals.max()

    fig = plt.figure(figsize=figsize, facecolor='white')
    ax_main = fig.add_axes([0.02, 0.02, 0.78, 0.90])
    ax_ak = fig.add_axes([0.02, 0.02, 0.22, 0.25])
```

```
    gdf_conus.plot(ax=ax_main, **plot_kw)
    ax_main.set_axis_off()
    ax_main.set_title(title, fontsize=20, fontweight='bold', pad=12)

    if not gdf_alaska.empty:
        gdf_alaska.plot(ax=ax_ak, **plot_kw)
    ax_ak.set_axis_off()

    # Colorbar - matplotlib handles tick placement natively for any norm
    cax = fig.add_axes([0.82, 0.08, 0.03, 0.74])
    sm = plt.cm.ScalarMappable(cmap=cmap, norm=norm)
    sm.set_array([])
    fig.colorbar(sm, cax=cax, label=cbar_label).ax.yaxis.label.set_fontsize(14)
    cax.tick_params(labelsize=14)

    if output_path:
        fig.savefig(output_path, dpi=dpi, bbox_inches='tight',␣
 ↪facecolor='white')
        print(f"  Saved: {output_path}")
    if show_plot:
        plt.show()
    else:
        plt.close(fig)
    return output_path if output_path else None


if gdf_conus is not None and gdf_alaska is not None:
    print("Generating choropleth maps...")

    # --- Break-even COP at 90% AFUE (green) ---
    create_choropleth_map(
        gdf_conus, gdf_alaska, column='breakeven_cop_90',
        title='Break-Even COP by State at 90% AFUE (2024)',
        cbar_label='Break-Even COP\n(COP needed for HP to match furnace cost)',
        output_path=os.path.join(PROJECT_ROOT, "state_breakeven_cop_90_map_2024.
 ↪png"),
        cmap='Greens', show_plot=True,
    )

    print("  Maps generated")
else:
    print("  Maps skipped - no geodataframe available")
```

```
Generating choropleth maps…
  Saved: c:\users\jorda\desktop\projects\cmu-tare-
model\state_breakeven_cop_90_map_2024.png
```




```
Maps generated
```

Display Results

[14]:
```
print("===== STATE PRICE RATIOS (2024 nominal) =====\n")
display(df_prices_csv)

for mp in selected_mps:
    print(f"\n===== THERMAL COP & AFUE (MP{mp}, NG homes) =====\n")
    display(cop_results[mp].sort_values('thermal_cop', ascending=False)[
        ['state', 'thermal_cop', 'baseline_afue', 'fans_pumps_pct',␣
 ↪'home_count']
    ])

# --- Comparison table: spark gap, break-even COP, effective COP ---
print(f"\n===== SPARK GAP & BREAK-EVEN COP vs EFFECTIVE COP (MP{primary_mp})␣
 ↪=====\n")

df_display = df_spark[[
    'state', 'state_name', 'spark_gap',
    'breakeven_cop_80', 'breakeven_cop_90', 'breakeven_cop_95',␣
 ↪'breakeven_cop_100',
    'thermal_cop', 'baseline_afue',
]].copy()
```

```
# Flag whether the heat pump beats the break-even threshold at 90% AFUE
df_display['hp_beats_breakeven_90'] = df_display['thermal_cop'] >␣
 ↪df_display['breakeven_cop_90']

display(df_display.sort_values('spark_gap', ascending=False).
 ↪reset_index(drop=True))
```

```
===== STATE PRICE RATIOS (2024 nominal) =====
```

	state	state_name	elec_price_kwh	gas_price_kwh	\
0	AK	Alaska	0.2482	0.0386	
1	MI	Michigan	0.1930	0.0354	
2	CT	Connecticut	0.2875	0.0553	
3	WI	Wisconsin	0.1718	0.0333	
4	CA	California	0.3197	0.0629	
5	NM	New Mexico	0.1420	0.0297	
6	IN	Indiana	0.1477	0.0312	
7	MT	Montana	0.1266	0.0270	
8	NY	New York	0.2443	0.0545	
9	NJ	New Jersey	0.1934	0.0434	
10	MN	Minnesota	0.1545	0.0350	
11	IL	Illinois	0.1587	0.0365	
12	CO	Colorado	0.1492	0.0348	
13	SD	South Dakota	0.1286	0.0306	
14	ID	Idaho	0.1152	0.0280	
15	MA	Massachusetts	0.2935	0.0720	
16	RI	Rhode Island	0.2865	0.0712	
17	PA	Pennsylvania	0.1777	0.0448	
18	IA	Iowa	0.1340	0.0340	
19	ME	Maine	0.2429	0.0623	
20	ND	North Dakota	0.1151	0.0296	
21	VT	Vermont	0.2190	0.0570	
22	NH	New Hampshire	0.2340	0.0616	
23	OH	Ohio	0.1599	0.0453	
24	TN	Tennessee	0.1242	0.0353	
25	DE	Delaware	0.1657	0.0486	
26	MD	Maryland	0.1786	0.0531	
27	NE	Nebraska	0.1153	0.0349	
28	KS	Kansas	0.1415	0.0432	
29	DC	District of Columbia	0.1771	0.0542	
30	WY	Wyoming	0.1247	0.0407	
31	WV	West Virginia	0.1507	0.0502	
32	KY	Kentucky	0.1279	0.0447	
33	WA	Washington	0.1190	0.0418	
34	VA	Virginia	0.1441	0.0506	
35	OR	Oregon	0.1470	0.0516	

36	UT	Utah	0.1222	0.0432
37	HI	Hawaii	0.4286	0.1607
38	OK	Oklahoma	0.1224	0.0460
39	MS	Mississippi	0.1339	0.0507
40	NC	North Carolina	0.1413	0.0540
41	SC	South Carolina	0.1423	0.0556
42	TX	Texas	0.1494	0.0591
43	AL	Alabama	0.1518	0.0612
44	NV	Nevada	0.1500	0.0614
45	GA	Georgia	0.1408	0.0598
46	MO	Missouri	0.1291	0.0554
47	AR	Arkansas	0.1232	0.0529
48	AZ	Arizona	0.1491	0.0673
49	LA	Louisiana	0.1173	0.0538
50	FL	Florida	0.1414	0.0834

	elec_price_mmbtu	gas_price_mmbtu	spark_gap
0	72.74	11.30	6.44
1	56.56	10.37	5.46
2	84.26	16.21	5.20
3	50.35	9.77	5.15
4	93.69	18.44	5.08
5	41.62	8.72	4.77
6	43.29	9.15	4.73
7	37.10	7.92	4.69
8	71.60	15.97	4.48
9	56.68	12.73	4.45
10	45.28	10.26	4.41
11	46.51	10.70	4.35
12	43.73	10.19	4.29
13	37.69	8.98	4.20
14	33.76	8.20	4.12
15	86.02	21.10	4.08
16	83.96	20.87	4.02
17	52.08	13.12	3.97
18	39.27	9.95	3.95
19	71.19	18.26	3.90
20	33.73	8.66	3.89
21	64.18	16.70	3.84
22	68.58	18.05	3.80
23	46.86	13.27	3.53
24	36.40	10.36	3.51
25	48.56	14.24	3.41
26	52.34	15.55	3.37
27	33.79	10.24	3.30
28	41.47	12.66	3.28
29	51.90	15.89	3.27
30	36.55	11.94	3.06

31	44.17	14.70	3.00
32	37.48	13.11	2.86
33	34.88	12.25	2.85
34	42.23	14.84	2.85
35	43.08	15.13	2.85
36	35.81	12.67	2.83
37	125.61	47.09	2.67
38	35.87	13.48	2.66
39	39.24	14.85	2.64
40	41.41	15.82	2.62
41	41.70	16.29	2.56
42	43.78	17.32	2.53
43	44.49	17.95	2.48
44	43.96	18.01	2.44
45	41.26	17.52	2.35
46	37.84	16.24	2.33
47	36.11	15.50	2.33
48	43.70	19.72	2.22
49	34.38	15.78	2.18
50	41.44	24.44	1.70

===== THERMAL COP & AFUE (MP3, NG homes) =====

	state	thermal_cop	baseline_afue	fans_pumps_pct	home_count
3	CA	3.054755	0.801801	4.001501	24571
8	FL	2.893052	0.794453	5.682569	1084
16	LA	2.758595	0.799742	5.761571	2139
41	TX	2.748806	0.804379	5.035227	12362
6	DC	2.715392	0.752035	4.055966	338
35	OR	2.618561	0.820170	3.725680	2183
0	AL	2.608147	0.805110	4.792433	1853
38	SC	2.603092	0.808995	4.670975	1580
9	GA	2.598895	0.812388	4.839350	5111
2	AZ	2.588321	0.823168	5.414082	2757
23	MS	2.584149	0.810445	5.177692	1219
45	WA	2.510266	0.817851	3.375780	3540
1	AR	2.499449	0.804689	5.161718	1582
34	OK	2.493255	0.806185	4.651168	2872
31	NV	2.491338	0.821705	4.966447	1998
25	NC	2.471322	0.807238	4.645363	3479
40	TN	2.407582	0.802505	4.752710	3023
18	MD	2.398846	0.789437	4.136133	3115
30	NM	2.384277	0.811805	5.304953	1506
43	VA	2.380144	0.801493	4.688277	3468
7	DE	2.314899	0.802174	4.076548	533
29	NJ	2.294246	0.795664	3.462339	6742
15	KY	2.290071	0.785014	4.005665	2261

22	MO	2.227515	0.795452	4.501767	4302
14	KS	2.200695	0.787260	3.997958	2599
42	UT	2.162580	0.821928	4.524524	2505
11	ID	2.154783	0.824162	4.403011	1118
5	CT	2.085552	0.803036	2.979604	1141
47	WV	2.051419	0.797393	2.992868	1076
36	PA	2.020353	0.795560	2.898380	8594
17	MA	2.012107	0.800042	2.691154	3090
33	OH	1.989596	0.803509	2.943619	10373
48	WY	1.977422	0.807016	4.351049	423
32	NY	1.974215	0.793466	2.439740	8652
13	IN	1.971601	0.804715	3.235824	5434
4	CO	1.952172	0.823187	4.292325	4566
12	IL	1.945797	0.804501	3.148780	11086
37	RI	1.936985	0.802848	2.434246	477
27	NE	1.897562	0.807015	3.166600	1507
24	MT	1.892323	0.809666	3.418760	705
28	NH	1.853915	0.819283	2.335242	232
20	MI	1.785815	0.809416	2.402143	9884
10	IA	1.763096	0.806999	3.002361	2637
44	VT	1.731443	0.805144	1.858951	110
46	WI	1.699153	0.804310	2.207530	4759
39	SD	1.673837	0.797856	2.435159	520
19	ME	1.647215	0.819947	1.661832	61
21	MN	1.589963	0.801746	1.937747	4687
26	ND	1.527465	0.803502	1.730071	402

===== THERMAL COP & AFUE (MP4, NG homes) =====

	state	thermal_cop	baseline_afue	fans_pumps_pct	home_count
3	CA	5.719542	0.801801	5.075078	24571
8	FL	5.697018	0.794453	5.459375	1084
16	LA	5.345519	0.799742	5.496742	2139
41	TX	5.303021	0.804379	5.558138	12362
0	AL	5.058210	0.805110	5.564781	1853
9	GA	5.057374	0.812388	5.699106	5111
38	SC	5.056874	0.808995	5.632238	1580
23	MS	5.024481	0.810445	5.541313	1219
6	DC	4.980642	0.752035	4.335234	338
1	AR	4.830257	0.804689	5.518244	1582
34	OK	4.799527	0.806185	5.577920	2872
25	NC	4.723897	0.807238	5.484151	3479
2	AZ	4.684796	0.823168	5.024505	2757
35	OR	4.661229	0.820170	5.222965	2183
40	TN	4.567381	0.802505	5.355325	3023
31	NV	4.481222	0.821705	5.013914	1998
43	VA	4.460033	0.801493	5.207943	3468

```
18    MD    4.428351       0.789437        4.937560        3115
7     DE    4.261080       0.802174        4.993519         533
45    WA    4.219903       0.817851        4.817898        3540
15    KY    4.146416       0.785014        4.782815        2261
29    NJ    4.130790       0.795664        4.734180        6742
30    NM    4.107143       0.811805        4.772070        1506
22    MO    4.083079       0.795452        5.001774        4302
14    KS    3.980038       0.787260        4.902813        2599
42    UT    3.697648       0.821928        4.869321        2505
5     CT    3.594760       0.803036        4.334917        1141
11    ID    3.594294       0.824162        4.885474        1118
47    WV    3.435717       0.797393        4.085933        1076
17    MA    3.374642       0.800042        4.013928        3090
33    OH    3.343245       0.803509        4.154727       10373
36    PA    3.341868       0.795560        3.909586        8594
13    IN    3.337385       0.804715        4.238571        5434
12    IL    3.301733       0.804501        4.245725       11086
32    NY    3.201888       0.793466        3.668657        8652
27    NE    3.179736       0.807015        4.236552        1507
37    RI    3.166801       0.802848        3.811201         477
4     CO    3.075468       0.823187        3.944798        4566
28    NH    3.055892       0.819283        3.936944         232
48    WY    2.960037       0.807016        3.671006         423
10    IA    2.849892       0.806999        3.731092        2637
24    MT    2.845549       0.809666        3.562685         705
20    MI    2.829020       0.809416        3.567197        9884
44    VT    2.676052       0.805144        3.111348         110
46    WI    2.623592       0.804310        3.174381        4759
39    SD    2.502694       0.797856        3.040386         520
19    ME    2.444452       0.819947        2.846700          61
21    MN    2.359931       0.801746        2.772684        4687
26    ND    2.149135       0.803502        2.371268         402

===== SPARK GAP & BREAK-EVEN COP vs EFFECTIVE COP (MP3) =====

   state            state_name  spark_gap  breakeven_cop_80  breakeven_cop_90  \
0     MI              Michigan       5.46              4.37              4.91
1     CT           Connecticut       5.20              4.16              4.68
2     WI             Wisconsin       5.15              4.12              4.64
3     CA            California       5.08              4.06              4.57
4     NM            New Mexico       4.77              3.82              4.29
5     IN               Indiana       4.73              3.78              4.26
6     MT               Montana       4.69              3.75              4.22
7     NY              New York       4.48              3.58              4.03
8     NJ            New Jersey       4.45              3.56              4.00
9     MN             Minnesota       4.41              3.53              3.97
10    IL              Illinois       4.35              3.48              3.91
```

11	CO	Colorado	4.29	3.43	3.86
12	SD	South Dakota	4.20	3.36	3.78
13	ID	Idaho	4.12	3.30	3.71
14	MA	Massachusetts	4.08	3.26	3.67
15	RI	Rhode Island	4.02	3.22	3.62
16	PA	Pennsylvania	3.97	3.18	3.57
17	IA	Iowa	3.95	3.16	3.56
18	ME	Maine	3.90	3.12	3.51
19	ND	North Dakota	3.89	3.11	3.50
20	VT	Vermont	3.84	3.07	3.46
21	NH	New Hampshire	3.80	3.04	3.42
22	OH	Ohio	3.53	2.82	3.18
23	TN	Tennessee	3.51	2.81	3.16
24	DE	Delaware	3.41	2.73	3.07
25	MD	Maryland	3.37	2.70	3.03
26	NE	Nebraska	3.30	2.64	2.97
27	KS	Kansas	3.28	2.62	2.95
28	DC	District of Columbia	3.27	2.62	2.94
29	WY	Wyoming	3.06	2.45	2.75
30	WV	West Virginia	3.00	2.40	2.70
31	KY	Kentucky	2.86	2.29	2.57
32	WA	Washington	2.85	2.28	2.56
33	VA	Virginia	2.85	2.28	2.56
34	OR	Oregon	2.85	2.28	2.56
35	UT	Utah	2.83	2.26	2.55
36	OK	Oklahoma	2.66	2.13	2.39
37	MS	Mississippi	2.64	2.11	2.38
38	NC	North Carolina	2.62	2.10	2.36
39	SC	South Carolina	2.56	2.05	2.30
40	TX	Texas	2.53	2.02	2.28
41	AL	Alabama	2.48	1.98	2.23
42	NV	Nevada	2.44	1.95	2.20
43	GA	Georgia	2.35	1.88	2.12
44	MO	Missouri	2.33	1.86	2.10
45	AR	Arkansas	2.33	1.86	2.10
46	AZ	Arizona	2.22	1.78	2.00
47	LA	Louisiana	2.18	1.74	1.96
48	FL	Florida	1.70	1.36	1.53

	breakeven_cop_95	breakeven_cop_100	thermal_cop	baseline_afue	\
0	5.19	5.46	1.785815	0.809416	
1	4.94	5.20	2.085552	0.803036	
2	4.89	5.15	1.699153	0.804310	
3	4.83	5.08	3.054755	0.801801	
4	4.53	4.77	2.384277	0.811805	
5	4.49	4.73	1.971601	0.804715	
6	4.46	4.69	1.892323	0.809666	
7	4.26	4.48	1.974215	0.793466	

8	4.23	4.45	2.294246	0.795664
9	4.19	4.41	1.589963	0.801746
10	4.13	4.35	1.945797	0.804501
11	4.08	4.29	1.952172	0.823187
12	3.99	4.20	1.673837	0.797856
13	3.91	4.12	2.154783	0.824162
14	3.88	4.08	2.012107	0.800042
15	3.82	4.02	1.936985	0.802848
16	3.77	3.97	2.020353	0.795560
17	3.75	3.95	1.763096	0.806999
18	3.70	3.90	1.647215	0.819947
19	3.70	3.89	1.527465	0.803502
20	3.65	3.84	1.731443	0.805144
21	3.61	3.80	1.853915	0.819283
22	3.35	3.53	1.989596	0.803509
23	3.33	3.51	2.407582	0.802505
24	3.24	3.41	2.314899	0.802174
25	3.20	3.37	2.398846	0.789437
26	3.14	3.30	1.897562	0.807015
27	3.12	3.28	2.200695	0.787260
28	3.11	3.27	2.715392	0.752035
29	2.91	3.06	1.977422	0.807016
30	2.85	3.00	2.051419	0.797393
31	2.72	2.86	2.290071	0.785014
32	2.71	2.85	2.510266	0.817851
33	2.71	2.85	2.380144	0.801493
34	2.71	2.85	2.618561	0.820170
35	2.69	2.83	2.162580	0.821928
36	2.53	2.66	2.493255	0.806185
37	2.51	2.64	2.584149	0.810445
38	2.49	2.62	2.471322	0.807238
39	2.43	2.56	2.603092	0.808995
40	2.40	2.53	2.748806	0.804379
41	2.36	2.48	2.608147	0.805110
42	2.32	2.44	2.491338	0.821705
43	2.23	2.35	2.598895	0.812388
44	2.21	2.33	2.227515	0.795452
45	2.21	2.33	2.499449	0.804689
46	2.11	2.22	2.588321	0.823168
47	2.07	2.18	2.758595	0.799742
48	1.62	1.70	2.893052	0.794453

	hp_beats_breakeven_90
0	False
1	False
2	False
3	False
4	False

5	False
6	False
7	False
8	False
9	False
10	False
11	False
12	False
13	False
14	False
15	False
16	False
17	False
18	False
19	False
20	False
21	False
22	False
23	False
24	False
25	False
26	False
27	False
28	False
29	False
30	False
31	False
32	False
33	False
34	True
35	False
36	True
37	True
38	True
39	True
40	True
41	True
42	True
43	True
44	True
45	True
46	True
47	True
48	True